

Myeloproliferative neoplasms — PV, ET, PMF

Philadelphia-negative MPNs driven by constitutive JAK-STAT activation (**JAK2 V617F**, **CALR**, **MPL**) · shared risks are **thrombosis**, progression to **myelofibrosis**, and **blast-phase transformation**

No Stanford handbook chapter — compiled from WHO 2022 / ICC criteria, NCCN MPN guidelines, and landmark trials. Verify against primary sources before acting.

DRIVER MUTATIONS

Driver	PV	ET	PMF
JAK2 V617F	~95%	50–60%	~60%
JAK2 exon 12	~3%	—	—
CALR	—	20–25%	25–30%
MPL	—	3–5%	5–8%
Triple-negative	—	10–15%	~10%

PV is essentially always JAK2-mutated — a JAK2-negative erythrocytosis should send you back to the differential, not to a PV diagnosis. In ET, CALR predicts higher platelets but **lower thrombosis** risk than JAK2.

MUTATION BIOLOGY

- **All three drivers converge on JAK-STAT** — JAK inhibitors work irrespective of which one is present
- **JAK2 V617F** — exon 14 **pseudokinase (JH2) domain**; relieves autoinhibition → ligand-independent signaling. **Homozygosity via 9p uniparental disomy** is common in PV/PMF, rare in ET; higher allele burden tracks with PV phenotype, pruritus and fibrotic progression
- **JAK2 exon 12** — ~3% of PV: **isolated erythrocytosis**, younger, very low EPO; outcomes similar to V617F
- **CALR** — exon 9 frameshift; the mutant C-terminus loses the **KDEL ER-retention motif** and instead activates **MPL** — which is exactly why CALR occurs in ET and PMF but **never in PV**. **Type 1 (del52)** predominates in PMF and is favorable; **type 2 (ins5)** in ET
- **MPL W515L/K** — exon 10 TPO receptor; ET and PMF only. **Triple-negative** — worst prognosis in PMF

DIAGNOSTIC ANCHORS (WHO 2022)

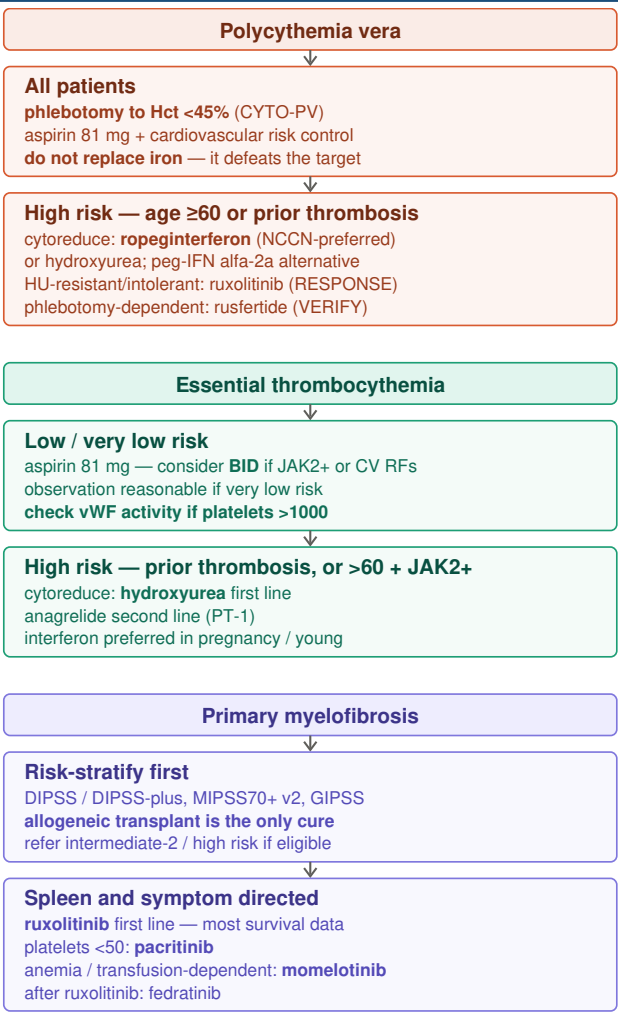
- **PV:** Hb >16.5 (M) / >16.0 (F) or Hct >49% (M) / >48% (F) + trilineage hypercellularity with pleomorphic megakaryocytes + JAK2 mutation; **subnormal EPO** is a minor criterion
- **ET:** platelets ≥450 ×10⁹/L + large hyperlobulated (“staghorn”) megakaryocytes + **reticulin ≤ grade 1** + a clonal marker, having excluded other MPNs
- **PMF:** megakaryocyte atypia with dense clustering + fibrosis grade (prefibrotic vs overt) + leukoerythroblastosis, anemia, splenomegaly, ↑LDH

The hardest and highest-stakes distinction is prefibrotic PMF vs ET. They look alike on counts, but prefibrotic PMF has materially worse survival and higher transformation risk. It turns on **marrow morphology** — megakaryocyte clustering and atypia — not on the CBC.

Masked PV: concurrent iron deficiency can normalize the hematocrit. If JAK2 is mutated with a “normal” Hct, think PV.

Splanchnic vein thrombosis (Budd-Chiari, portal vein) — test JAK2 in everyone, even with normal counts; MPN is a leading occult cause.

TREATMENT PATHWAY



RISK STRATIFICATION

Disease	Scheme
PV	High risk = age ≥60 OR prior thrombosis ; low risk = neither. High risk (or symptomatic/progressive) triggers cytoreduction
ET	IPSET-thrombosis (revised) : very low = ≤60, no thrombosis, JAK2 WT · low = ≤60, no thrombosis, JAK2-mutant · intermediate = >60, no thrombosis, JAK2 WT · high = prior thrombosis, or >60 AND JAK2-mutant
PMF	IPSS (at diagnosis) · DIPSS / DIPSS-plus (any time) · MIPSS70 / MIPSS70+ v2 (molecular; transplant candidates) · GIPSS (genetic only)

Non-driver mutations set prognosis, not diagnosis. High-molecular-risk in PMF: **ASXL1**, **SRSF2**, **EZH2**, **IDH1/2**, **U2AF1 Q157** — these drive the MIPSS scores and push toward transplant. **TP53** drives blast-phase transformation.

AGENTS & DOSING

Agent	Dose	Setting
Aspirin	81 mg daily; BID if JAK2+ or high-risk ET	all PV, most ET
Hydroxyurea	500–1000 mg daily, titrated to counts	1st-line cytoreduction
Ropeginterferon alfa-2b	100 µg SQ q2wk (50 if switching from HU); titrate by 50 to max 500	NCCN-preferred 1st-line PV
Peginterferon alfa-2a	45–180 µg SQ weekly	alternative; pregnancy
Rusfertide (Mimrylo)	weekly SQ, dose-titrated	PV erythrocytosis, phlebotomy-dependent
Anagrelide	0.5 mg BID, titrate	2nd-line ET
Ruxolitinib	PV 10 mg BID · MF 5–20 mg BID by platelet count	HU-resistant PV; 1st-line MF
Fedratinib	400 mg daily	MF after ruxolitinib
Pacritinib	200 mg BID	MF, platelets <50
Momelotinib	200 mg daily	MF with anemia (also anti-ACVR1)

Fedratinib requires thiamine — check and replete before and during treatment (Wernicke encephalopathy). **Never stop ruxolitinib abruptly** — cytokine rebound/withdrawal syndrome; taper.

COMPLICATIONS & PEARLS

- **Acquired von Willebrand syndrome** with extreme thrombocytosis (>1000–1500) — paradoxical **bleeding**. Check ristocetin cofactor before starting aspirin; hold aspirin if severely reduced
- **CYTO-PV:** Hct target <45% gave ~4× fewer cardiovascular deaths and major thromboses than a 45–50% target — the number to defend
- **Post-PV / post-ET myelofibrosis** (IWG-MRT criteria) — suspect with rising LDH, falling counts, enlarging spleen, new constitutional symptoms
- **Blast phase (≥20% blasts)** is dismal and chemo-refractory — allogeneic transplant is the only meaningful option
- **Aquagenic pruritus** in PV — antihistamines, SSRIs, interferon, or ruxolitinib (a genuine quality-of-life issue, not a nuisance)
- Anemia in MF: ESAs if EPO <500, danazol, luspatercept, or switch to momelotinib